

Orlando Joaqui Barandica

PHD

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About me

- I'm a Ph.D. in Engineering with a focus on Industrial Engineering from Universidad del Valle, where I also earned my degrees in Statistics and Applied Economics. Currently, I'm an assistant professor at Universidad del Valle, where I lead courses in statistics, econometrics, business analytics, machine learning for finance, data visualization, and quantitative finance. Throughout my academic journey, I've collaborated with several leading universities in Colombia, including Pontificia Universidad Javeriana Cali, Universidad ICESI, Universidad San Buenaventura Cali, and Universidad del Tolima—sharing knowledge and training students in areas at the intersection of data and finance. My research explores the dynamics of energy markets and energy finance, as well as quantitative approaches in business analytics, risk management, and econometrics. I've published in specialized journals on energy and finance, contributing to the development of data-driven insights that inform policy and business decisions in complex, evolving environments. More info: www.joaquibarandica.com

RESEARCH INTERESTS

- Energy Finance
- Business Analytics
- Quantitative Finance
- Applied Econometrics and Statistics
- Energy Markets

Education

PhD in Industrial Engineering

Cali, Colombia

UNIVERSIDAD DEL VALLE

2020 - 2023

- Research: Topics in Asset and Liability Management: a comparative vision between emerging and developed economies
- Advisors: Diego F. Manotas - Jorge M. Uribe

MSc Applied Economics

Cali, Colombia

UNIVERSIDAD DEL VALLE

2015 - 2017

- Research: Effect of the world interest rate on the equity markets of emerging economies
- Advisors: Jorge M. Uribe

BSc Statistics

Cali, Colombia

UNIVERSIDAD DEL VALLE

2008 - 2014

- Research: Modeling the effect of climatological factors on the productivity of rice cultivation in some producing areas of Colombia with information from historical series
- Partner: Oscar W. Orozco
- Advisors: David Arango - Javier Olaya

COMPLEMENTARY COURSES

Jun. 2018 **Big Data (Text Mining)**: (Cali, Colombia)

Pontificia Universidad Javeriana de Cali

May 2017 **Introduction to Mixed Generalized Linear Models**: (Cali, Colombia)

Colombian Society of Statistics

Apr. 2016 **Data Mining**: (Cali, Colombia)

Colombian Society of Statistics

Work Experience

Universidad del Valle

Cali, Colombia

ASSISTANT PROFESSOR

2016 - Current

- School of Statistics
- School of Industrial Engineering
- Department of Economics (2020)

Pontificia Universidad Javeriana de Cali

Cali, Colombia

ASSISTANT PROFESSOR

2016 - 2025

- Department of Accounting and Finance
- Department of Economics
- Department of Natural Sciences and Mathematics (2016 - 2019)

Universidad del Tolima

Ibagué, Colombia

ASSISTANT PROFESSOR

2021

- Department of Economics and Finance

Universidad ICESI

Cali, Colombia

ASSISTANT PROFESSOR

2016 - 2019

- Department of Mathematics and Statistics

Universidad San Buenaventura Cali

Cali, Colombia

ASSISTANT PROFESSOR

2019

- Department of Economics

Universidad Santiago de Cali

Cali, Colombia

ASSISTANT PROFESSOR

2018

- Education Faculty

CIAT - The International Center for Tropical Agriculture

Palmira, Colombia

VISITING RESEARCH

2014

- Program cassava breeding

Publications

ARTICLES

2026

- **Financial resilience of electricity sector companies in emerging and developed economies: a comparative analysis during times of distress.** (with Manotas-Duque, D. F., Orozco-Cerón, O.). *Quantitative Finance and Economics*. InPress

2025

- **Evaluation of Energy Complementarity in Colombia: An Analysis of Climate Variability and Non-Conventional Sources.** (with Manotas-Duque, D. F., Rivera-Cadavid, L.). *Results in Engineering*.
<https://doi.org/10.1016/j.rineng.2025.105824>
- **Hybrid VAR-XGBoost Modeling for Data-Driven Forecasting of Electricity Tariffs in Energy Systems Under Macroeconomic Uncertainty.** (with López-Estrada, S., Orozco-Cerón, O. W.). *Technologies*.
<https://doi.org/10.3390/technologies13110495>
- **Integrating Equity into Energy Efficiency Assessment: A Metafrontier Malmquist-Luenberger Analysis of Energy Poverty.** (with López-Estrada, S., Heredia-Carroza, J.). *Environment, Development and Sustainability*.
<https://doi.org/10.1007/s10668-025-06889-8>
- **Interaction between Armed Conflicts and Energy Prices in Colombia.** (with Manotas-Duque, D. F., López-Estrada, S.). *International Journal of Energy Economics and Policy*.
<https://doi.org/10.32479/ijeep.16887>
- **Macroeconomic Drivers of Financial Performance in Power Generation Firms across Emerging and Developed Markets.** (with Muñoz-Alzate, L. F., Manotas-Duque, D. F.). *Entramado*.
<https://doi.org/10.18041/1900-3803/entramado.2.12885>
- **Mapping Trends in Green Finance: A Bibliometric and Topic Modeling Analysis.** (with Heredia-Carroza, J., López-Estrada, S., Agheorghiesei, D.-T.). *International Journal of Financial Studies*.
<https://doi.org/10.3390/ijfs13030137>
- **Modeling the Impact of G7 Interest Rates on BRICS Equity Markets: A DLNM Approach Using MSCI Indices.** (with Heredia-Carroza, J., López-Estrada, S., Agheorghiesei, D.-T.). *Economies*.
<https://doi.org/10.3390/economies13090252>

- **Pathways to Specialized Renewable Energy Generation: Insights from Integer Portfolio Optimization in a Globalized Electricity Market.** (with Manotas-Duque, D. F., Uribe, J. M.). *Energy, Sustainability and Society*.
<https://doi.org/10.1186/s13705-024-00508-6>
- **The Sovereign Risk Amplifies ESG Market Extremes: A Quantile-Based Factor Analysis.** (with Orozco-Cerón, O. W., Manotas-Duque, D. F.). *Risks*.
<https://doi.org/10.3390/risks13120245>
- **Using Markov Chains and Entropy to Explain Value at Risk in European Electricity Markets.** (with Orozco-Cerón, O. W., Manotas-Duque, D. F.). *Journal of Risk and Financial Management*.
<https://doi.org/10.3390/jrfm18100591>

2024

- **Common Factors in the Profitability of Energy Firms.** (with Manotas-Duque, D. F., Uribe, J. M.). *The Energy Journal*.
<https://doi.org/10.1177/01956574241280779>
- **Exploring the Asymmetric Relationship between Macroeconomic Factors and Corporate Profitability in the MSCI Colombia Index.** (with Osorio-Vanegas, B., Ramirez-Patiño, C., Ojeda-Echeverry, C.). *Journal of Economics, Finance and Administrative Science*.
<https://doi.org/10.1108/JEFAS-08-2023-0234>
- **Macrofinancial Interconnections in the Pacific Alliance: A Quantile Approach of Stock Markets and Macroeconomic Factors.** (with Gómez-Daza, J. A., López-Estrada, S.). *Applied Economics Letters*.
<https://doi.org/10.1080/13504851.2024.2383259>
- **Non-Linear Dynamics of Global Liquidity and Energy Sector Profitability.** (with Manotas-Duque, D. F.). *Energy Sources, Part B: Economics, Planning and Policy*.
<https://doi.org/10.1080/15567249.2024.2407772>
- **Weather Conditions, Climate Change, and the Price of Electricity.** (with Mosquera-López, S., Uribe, J. M.). *Energy Economics*.
<https://doi.org/10.1016/j.eneco.2024.107789>

2023

- **Directional Predictability between Interest Rates and the Stoxx 600 Banks Index: A Quantile Approach.** (with Oviedo-Gomez, A., Manotas-Duque, D. F.). *Finance Research Letters*.
<https://doi.org/10.1016/j.frl.2023.104328>
- **How Do Climate and Macroeconomic Factors Affect the Profitability of the Energy Sector?.** (with Manotas-Duque, D. F.). *International Journal of Energy Economics and Policy*.
<https://doi.org/10.32479/ijeeep.14303>
- **Nonlinear Predictive Relationship Between GDP per capita and Mortality Rate: UK Case Study.** (with Orozco-Cerón, O. W.). *Desarrollo y Sociedad*.
<https://doi.org/10.13043/DYS.93.5>

2022

- **Assets Liability Management: A Bibliometric Analysis and Topic Modeling.** (with Manotas-Duque, D. F.). *Entramado*.
<https://doi.org/10.18041/1900-3803/entramado.1.8242>
- **Commonality, Macroeconomic Factors and Banking Profitability.** (with Manotas-Duque, D. F., Uribe, J. M.). *The North American Journal of Economics and Finance*.
<https://doi.org/10.1016/j.najef.2022.101714>
- **Mind the Gap! Socioeconomic Determinants of the Stunting Urban-Rural Gap for Children in Colombia.** (with Cárdenas, E., Osorio, A., Pico, S.). *Child Indicators Research*.
<https://doi.org/10.1007/s12187-021-09880-7>

2016

- **Cassava Breeding I: The Value of Breeding Value.** (with Ceballos, H., Pérez, J., Lenis, J., Morante, N., Calle, F., Pino, L., Hershey, C.). *Frontiers in Plant Science*.
<https://doi.org/10.3389/fpls.2016.01227>

– **Cassava Breeding II: Phenotypic Correlations through the Different Stages of Selection.** (with Pérez, J., Lenis, J., Calle, F., Morante, N., Pino, L., Hershey, C., Ceballos, H.). *Frontiers in Plant Science*.
<https://doi.org/10.3389/fpls.2016.01649>

RESEARCH IN PROGRESS

- Under Review **Cross-Quantile Dynamics of China's Energy Investment and Commodity Prices.** . .
- Under Review **Climate-stress risk propagation in a hydro-dominated power system: quantile connectedness and CoVaR evidence from Colombia.** . .
- Under Review **Quantile Connectedness between Energy Transition and Tourism Dynamics: Evidence from Spain (2019–2024).** . .
- Under Review **Social and Environmental Drivers Shaping Waste Source Separation Behavior in Higher Education.** . .
- Under Review **Entrepreneurial and Civil Service Career Intentions among the Young Population in Rural Areas: A Theory of Planned Behaviour Analysis.** . .
- Under Review **Daily Biomass Electricity as a Bioeconomy Flexibility Service under Climate Stress: GAM Evidence from Colombia.** . .

CHAPTER OF THE BOOK

- 2023 **Fondos de inversión colectiva (FIC) como determinantes de la estabilidad del mercado accionario en Colombia.** (with Gómez-Daza, Jesús A., Barrios-Álvarez, Claudia). *Las ciencias económicas toman la palabra: Diálogos en un mundo cambiante*. Ed. Sello Editorial Javeriano. ISBN: 9786287709034
- 2019 **Resultados componente cuantitativo: acciones, conocimientos, actitudes y prácticas preventivas frente al VIH.** (with Canaval, G., Valencia, C., Muñoz, E., Tovar, R., Triviño, Z. Chantre, S., Chamorro, M.). *Tramas, sexualidades y prevención del VIH en jóvenes universitarios de Cali*. Ed. Sello Editorial Javeriano. ISBN: 78-958-5119-18-5

Conferences

- July 2024 **Armed Conflict Dynamics and Energy Price Volatility: A Quantile Connectivity Approach.** (with López-Estrada Sebastián). 9 ELAEE. (Rio de Janeiro, Brazil).
- July 2024 **Assessment of Energy Price Dynamics: A GAS Model and Quantile Regression Analysis between Natural Gas, Climate, and Electricity Markets.** (with Hernández-Rosero, W., Ramirez, G., Ojeda, C.). 9 ELAEE. (Rio de Janeiro, Brazil).
- Dic. 2022 **Decision tools from index fund finance to explore the path towards a scenario of renewable energy generation with globalization and high specialization of regional electricity markets.** (with Manotas-Duque, Diego F., Uribe, Jorge M.). CLAIO 2022 – XXI Latin Ibero-American Conference on Operations Research. (Buenos Aires, Argentina).
- Nov. 2022 **Decision tools from index fund finance to explore the path towards a scenario of renewable energy generation with globalization and high specialization of regional electricity markets.** (with Manotas-Duque, Diego F., Uribe, Jorge M.). ELAEE 2022 – Energy Transition in Latin America. (Bogotá, Colombia).
- Jun. 2022 **Global liquidity and the profitability of energy firms.** (with Manotas-Duque, Diego F., Uribe, Jorge M.). ICEE 2022 – Energy and Environment: Bringing together Economics and Engineering. (Porto, Portugal).
- Sept. 2021 **Commonality, macroeconomic factors and banking profitability.** (with Manotas-Duque, Diego F., Uribe, Jorge M.). IFABS 2021. (Oxford, UK - (virtual)).

- May 2019 **Social Responsibility and Financial Performance in violent settings: A study of Colombian Family Firms.** (with González, A., Rodríguez, Y., Maldonado, S., Pérez, M.). 15th Annual Family Enterprise Research Conference – FERC.
(The University of Vermont, USA.).
- Oct. 2016 **Factores determinantes de la aplicación de los niveles de la GRI en la divulgación de información sobre responsabilidad social empresarial (RSE) para los años 2011 a 2015.** (with Maldonado, S., Gaitán, C., Acosta, A.). International Finance Conference - IFC. ISBN 978-956-362-941-5
(Viña del Mar, Chile).

Refereeing

JOURNALS

- Journal **Desarrollo y Sociedad**
Universidad de los Andes
- Journal **Entramados**
Universidad Libre de Cali
- Journal **Finance Research Letters**
Elsevier
- Journal **Annals of Operation Research**
Springer
- Journal **Applied Economics Letters**
Taylor and Francis
- Journal **Energy**
Elsevier
- Journal **Journal of Economics Finance and Administrative Science**
Emerald Publishing
- Journal **Journal of Infrastructure, Policy and Development**
EnPress Publishing
- Journal **Scientific Reports**
Nature

Software

Language R, Python